Healthcare Analytics For Doctor Visits TIRTC

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PROBLEM STATEMENT

- **Rare, uneven use:** About 80% of surveyed people made no doctor visit in two weeks, while a small group made many.
- **Overlapping factors:** Age, income, health and insurance are tangled together, so raw comparisons can mislead.
- **Key question:** Which factors truly drive visits once medical need is accounted for, and where are the access gaps?



Project Description

- **Data:** 5,190 survey respondents with demographics, health status and insurance type; target is doctor visits in the past 2 weeks.
- **Prepare:** Encoded flags, converted age and income to natural units, and built age, insurance and health-band groups. No missing values.
- **Explore and test:** Univariate, bivariate and multivariate EDA, then Mann-Whitney, Kruskal-Wallis, chi-square and Spearman tests.
- **Model:** Poisson vs Negative Binomial count models, comparing raw and adjusted effects to separate medical need from access.

WHO ARE THE END USERS?

- **Hospital and clinic managers:** plan capacity around patients with high medical need.
- **Health insurers and policymakers:** spot access gaps, such as low visit rates under low-income government cover.
- **Care-coordination teams:** target outreach at the under 2% of people who account for about 28% of all visits.
- **Health data analysts:** reuse the count-model approach for utilisation data.

Technology Used

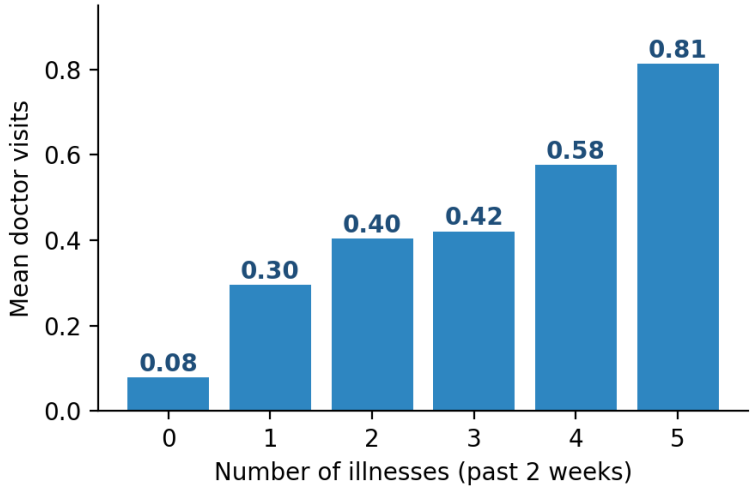
- **Python and Jupyter Notebook:** end-to-end analysis workflow
- **pandas and NumPy:** cleaning, feature engineering, grouping
- **Matplotlib and Seaborn:** charts and heatmaps
- **SciPy:** Mann-Whitney, Kruskal-Wallis, chi-square, Spearman tests
- **Statsmodels:** Poisson and Negative Binomial count regression



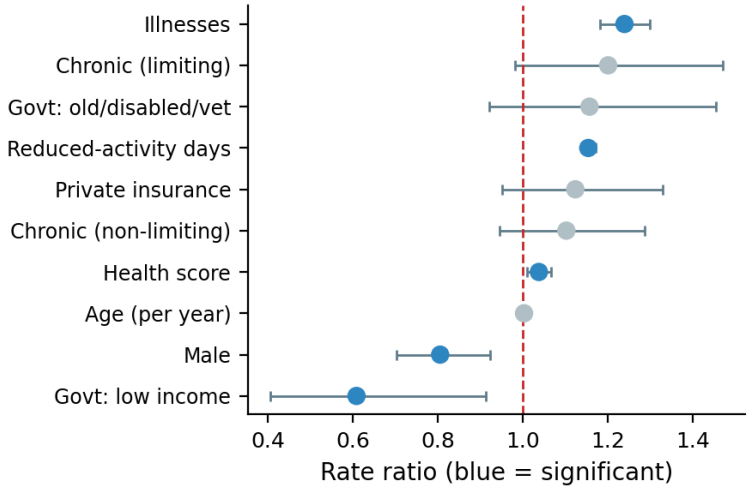
RESULTS

Illness, reduced activity and poor health drive visits; men and low-income government cover visit less than health predicts.

Visits rise with illness count



Adjusted effect on visits



GitHub: https://github.com/aakashpatil8641-gif/Healthcare-Analytics-for-Doctor-Visits

Thank you